ATRC ADVANCED TECHNOLOGY PIONEERS 2026 · ENEC TRACK

QANDEEL

قنديل

إدارة أسراب قناديل البحر وإعادتها إلى البحر حيّة

Herd, don’t harvest: autonomous, live-release management of Blue Blubber jellyfish swarms after ENEC’s early warning

Team overview · Team Qandeel

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EXECUTIVE SUMMARY

We keep swarms out of the intake and return them to the sea alive

Problem

Blue Blubber swarms (30–45 cm bells) drift onto seawater intakes. Clearing them takes heavy manual work and almost none survive.

Solution

After ENEC's alert, a drift planner decides when to act; a bubble curtain at the breakwater gap holds the swarm out; two uncrewed boats tow a boom with a closed-bottom bag and release the jellyfish alive. An AI gap camera is the backup.

Evidence

Simulation on real 2025 Gulf currents and wind, stress-tested against physics the planner does not know; online dashboard; 25 tests; open code.

Next

Recalibrate the model with ENEC's alert history, screen logs, flow and depth data, then a one-section sea pilot.

~3% vs ~23%

of arriving jellyfish reach the gap before the curtain runs: Qandeel vs a daylight camera alone (stress test on real 2025 data; mean of 3 × 100 alerts, range 2.7–3.7% vs 20–26%)

1–2 days

of warning for the boats and the curtain, for ~12 extra curtain hours per alert vs camera-only (~USD 500 of extra energy)

84–100%

of jellyfish kept by the boom with our retention bag (model), vs 0–55% for an open oil-spill boom

29–46%

of the energy of a fixed airflow, by setting air hour by hour from real currents and wind (variable-speed compressor)

THE PROBLEM AND WHY IT MATTERS

A large, local, hard-to-handle jellyfish

ENEC already detects swarms offshore. The open question is what to do next: manage high volumes with less manual work, less harm and low energy, in line with the UAE Net Zero 2050 and biodiversity goals.

The target: Blue Blubber

- One of the two dominant jellyfish on the UAE coast in 2019–2022 surveys (Aravind 2024, UAEU)
- 30–45 cm bell (ENEC brief; SeaLifeBase max ~35 cm); no published weight: ~1–3 kg across 30–45 cm by size scaling; we use 2 kg
- Lives near the surface of bays and estuaries, ~0–1 m (SeaLifeBase)
- Small adults swim ~4 cm/s (lab data); large adults unmeasured, so we test 10 and 20 cm/s

What happens at intakes

- Swarms follow the intake flow and pile on the screens, cutting cooling water
- Worldwide this has forced power reductions and shutdowns (e.g. Torness 2011, Oskarshamn 2013) (press reports)
- In the region, ~300 t of jellyfish clogged intakes in Oman (Gulf News, Ghubra); at Kuwait's Al-Zour plant (2017) they blocked the screens for days (Jamal et al., PAAET)

Options today, compared

OPTION	JELLYFISH	DRAWBACK
Rake screens (now)	die	reactive, heavy labour, outage risk
Fixed nets offshore	die, bycatch	clog in big blooms
Cutting robot (JEROS)	shredded	~0.9 t/h, kills
Water jets + LiDAR/drones (Kuwait, PAAET)	pushed away	pumps run the jets at the intake
Harvest for food	used	no UAE market yet
Qandeel	**released alive**	unproven at sea: pilot first

We found no published management trial on this species.

OUR SOLUTION · SYSTEM DESIGN

Guard the mouth, move the water, keep them alive

ENEC alert
swarm position
(existing system)

→

Qandeel planner
- chance of arrival and window
- curtain switch-on time
- airflow, crew side, release point
- live current and wind forecast

→

Bubble curtain
at the intake mouth,
only when needed

→

Uncrewed boats + boom + bag
gather in the water

→

Live release
down-current

AI gap camera + pressure sensors feed back



1 · The mouth is a chokepoint

Seawater reaches the pumps through one ~3 km channel; its mouth measures 967 m on satellite images.

2 · Move the water, not the animal

A rising-bubble current holds jellyfish out at sea. Nothing is cut or netted, and unlike jets or pumps nothing moves underwater: the compressor is on land. Ten ~97 m sections; only those in the swarm's path run.

3 · Proven tools, made autonomous

Oil-spill booms on uncrewed boats, plus a bag that stops leaks. Release west, away from the discharge.

One swarm, from alert to live release

1. **Alert.**
   ENEC reports a swarm, e.g. 15 km NNW. The planner receives it automatically.
2. **Plan.**
   400 simulated drifts of 60 swimming jellyfish each, on the live current and wind forecast: 44% chance of arrival, window 16–73 h.
3. **Prepare.**
   Curtain on at +13 h; uncrewed boats staged on the approach side; release point chosen.
4. **Deflect.**
   Curtain across the gap holds the swarm outside; airflow follows the current.
5. **Gather.**
   Two uncrewed surface vessels, supervised from shore, close a U-boom with a closed-bottom bag.
6. **Release.**
   Tow ~3 km NNW (≤0.2 m/s through the water: ~7 h over the ground, up to a day against strong inflow) and open the bag; ~60% drift back and are held again.
7. **Learn.**
   An AI camera counts jellyfish at the gap and starts the curtain if they arrive early. Its records are the training data for a learning model that corrects the drift forecast.

44%
chance the swarm reaches the gap

16–73 h
arrival window after the alert

+13 h
curtain switch-on

3 km NNW
release point (~7 h tow)



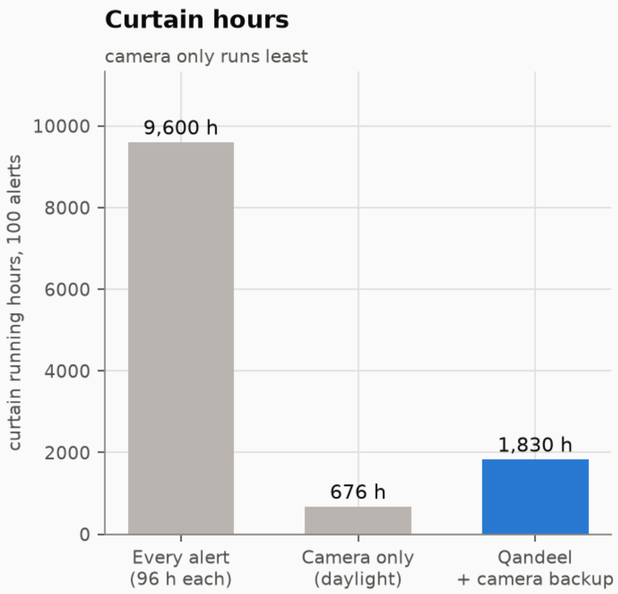
From the live dashboard, alert 15 km NNW: swarm cloud, release point, curtain status. Data: Copernicus SMOC + ERA5, summer 2025.

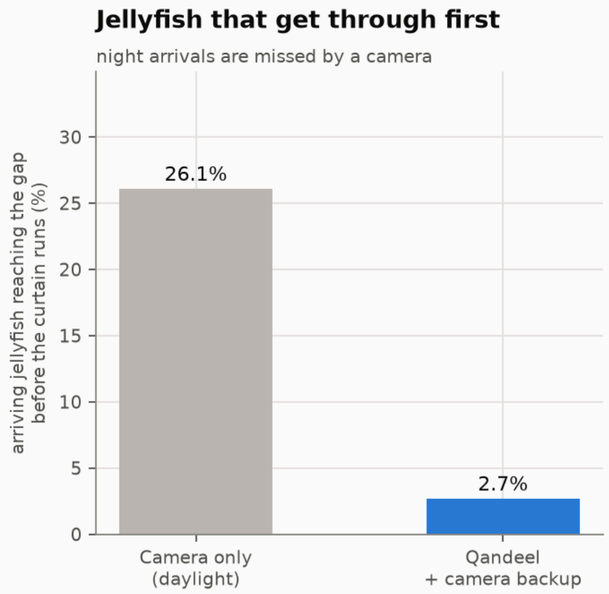
How the plan is calculated

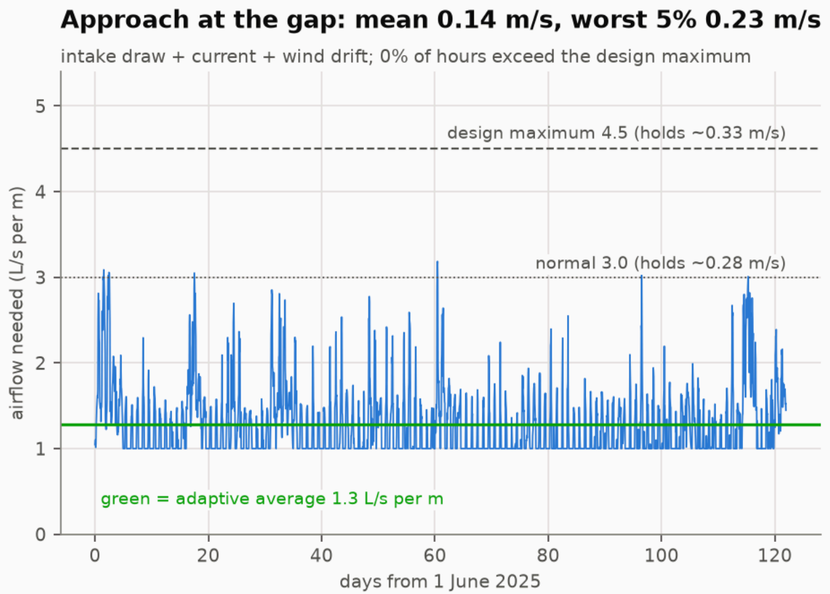
Arrival: 400 runs × 60 virtual jellyfish moved every 10 min by current, 1% wind drift, swimming and mixing; chance = runs where 5% come within 2 km.
Switch-on: if chance ≥10%, earliest likely arrival − 3 h; off once the window has passed and the camera sees none for 6 h.
Release: of 50+ points west to NNE, the one removing most jellyfish per boat-hour, avoiding strandings near the plant.
Airflow: set hourly to beat the approach current with a margin.

Better protection than a camera alone, at a known cost

Stress test, real 2025 season: Qandeel lets ~3% through vs ~23% for a camera alone (mean of 3 runs; range 2.7-3.7% vs 20-26%)







100 alerts at random times and places in the real 2025 season; the truth uses physics the planner does not know. Base case 1% wind drift. “Arrived” = within 2 km of the gap (15 km NNW alert: 44%; within 500 m: 31%). Bars show one run (2.7% vs 26.1%); the card gives the mean of three.

~3% vs ~23% get through

Truth: currents ×0.7–1.3, wind drift ×0.5–1.7, plus a 0–3 cm/s current the planner doesn’t know. A daylight camera misses night arrivals.

The cost

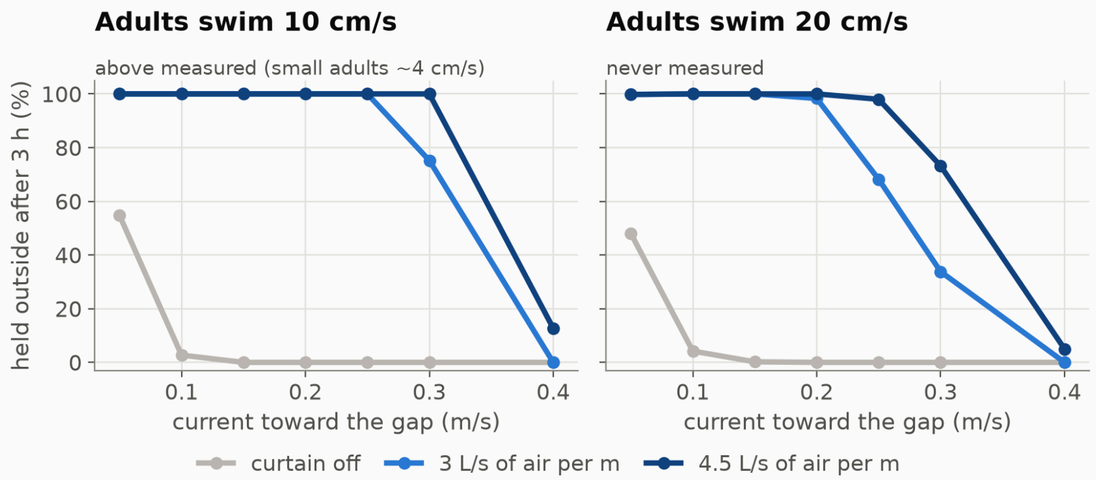
~1,800 curtain hours vs ~700 camera-only: ~12 h and ~USD 500 more energy per alert; ~80% less than running on every alert. In return: 1–2 days of warning.

What this is not

Still simulation against simulation, not real jellyfish. Next: compare drift with real Gulf drifter tracks and ENEC swarm records.

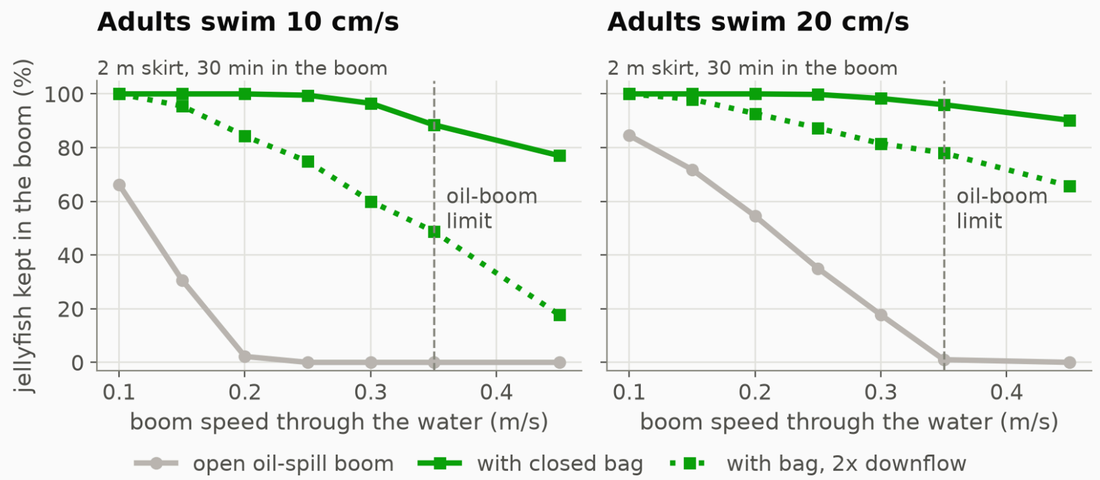
Oil-spill booms leak jellyfish; a closed bag fixes it

More air holds against faster water



Each point: 400 simulated jellyfish. Bubble-driven surface current from the standard bubble-curtain formula (Bulson 1968: current grows with the cube root of airflow); its constant is confirmed in the pilot.

Open booms leak jellyfish; the bag keeps them



A towed boom pulls water down behind it (“downflow”): we assume 25% of tow speed, dotted line 50%. Jellyfish depths from field data.

Curtain holds, if adults swim slowly

Water approaches the gap at 0.14 m/s on average, 0.23 m/s in the worst 5% of hours (0.20 / 0.36 if wind drift is 3%). 3 L/s per m holds ~0.27 m/s; 4.5 holds ~0.33. Over a full 82 h event: 100% held at 0.2 m/s, but 78% if adults swim 20 cm/s.

Why the bag matters

Oil floats back up; jellyfish drift with the water and are dragged under the skirt well below the 0.35 m/s oil-boom limit (2–55% kept at 0.2 m/s). A closed-bottom fabric bag keeps 84–100% in the model.

Where held jellyfish go. Boats clear only ~0.4–1.4 t/h, so a 300 t bloom (Oman) would take them 9–30 days: they don’t remove the swarm, they clear what piles up at the mouth. The curtain holds the swarm while the sea moves it on: in our 2025 data the flow runs along the coast at ~0.07 m/s (~6 km a day) and turns offshore 22% of the time (onshore spells last a median 14 h). The curtain stays on until the planner’s window has passed and the camera has seen no jellyfish at the mouth for 6 h, so a held swarm cannot pour in when it stops.

Gentle on marine life, built from parts that exist today

Marine-life protection

RISK	HOW WE REDUCE IT
Bubbles drawn into the intake	Curtain on the seaward side; bubbles surface within a minute, in the outer basin; bubble detector on the intake line; curtain moved out if the pilot shows carry-over
Nuclear approval, security	FANR approval from day one; kit stays outside safety-related intake structures; boats run by plant staff under security-zone rules
Jellyfish harmed	Nothing cuts, pumps or nets them: the curtain's surface current pushes them back. Effect of bubbles on the bell not yet known: checked in the pilot
Turtles, dolphins, dugongs, seagrass	Escape gap, slow tow, camera stop on sighting; pipe routed away from seagrass and coral; release points kept out of protected areas
Crowding in the bag during the tow	Bag stays underwater, where jellyfish are nearly weightless, and seawater flows through the fabric: we expect far less crushing than in hauled nets. At most half full; opened at sea by 12 h. Survival is a pilot pass mark
Released jellyfish strand	Left alone, 55% of this swarm strands within 4 days; released 3 km out, 29% do, most still within 10 km of the plant. Release points refined with EAD
Sediment, bubbly water	Diffuser pipe on legs 0.5–1 m above the bed, so sand cannot bury the holes and block the air (less repair work); turbidity watched; no boats or divers in the bubble zone while it runs

Feasibility

ITEM	ESTIMATE (967 M MOUTH)
Compressed air	2.9 m^3/s, ~440 kW for the 967 m mouth, oil-free (ISO 8573-1 Class 0)
Capital (indicative)	USD 1.2–3.5 M: 967 m curtain, variable-speed compressor + standby, boom, bags, two uncrewed boats, sensors
Running cost	USD 0.18–0.45 M a year (operators, boat upkeep, diver cleaning, servicing). A full 72 h event: ~31 MWh, ~USD 3,100; 12.6 t CO_2 on a gas grid, 0.4 t on the plant's own power
Value	Torness (UK) lost an estimated ~USD 1.5 M (~£1 M) a day while jellyfish kept it shut in 2011 (press estimate): avoiding 0.8–2.3 such days repays the capital
Capacity	Gathering 4–43 t/h, but the loaded tow sets the pace: ~0.4 t/h per boom unit, ~1.4 t/h with three uncrewed tugs swapping bags. The curtain does the bulk work; boats clear what builds up
Site works	Compressor house on land at the breakwater root; air main with power and fibre along the ~3 km breakwater; valve chamber and camera mast at the tip; boat base inside the channel
Gulf hardening	Membrane diffusers, foul-release coating, modular sections, lifted out off-season

ASSUMPTIONS (NO DATASET PROVIDED) AND LIMITATIONS

What we assume, and what we measure first

ASSUMPTION	BASIS	IF WRONG
Intake mouth 967 m wide (measured), ~9.5 m deep	Satellite images; public channel depth	Depth to survey; a shorter curtain inside the channel would cut air and cost
Alert gives position, hours of notice	Brief: “time to prepare”	Camera trigger switches the curtain on
Swarm in top 1 m; adults swim ≤10–20 cm/s	Field and lab studies (Australia)	Deeper skirt; tested at 20 cm/s
Swarm density 0.1–1 per m^3	No published figure	Capacity scales linearly
Extra wind drift 1% of wind (0–3%)	Ocean model already has wind-driven drift	28 km N: 3–29% arrival. Curtain sized for 1%; at 3%, ~8% of hours exceed its max (flagged)
Apex downflow 25–50% of tow speed; bag passes 20%	Model hypothesis	Measured in the sea pilot; bag tested before scale-up

Limitations: tested only against simulations; currents from one model point, rounded to ~0.03 m/s by the free API (direct Copernicus download ready); no depth: the curtain draws water in near the bed, so deep or night-time jellyfish could pass under; species at Barakah and adult weight to confirm. **Why not harvest?** A full bag can go to a licensed processor; release is the default while no UAE market exists.

What moves the results most

One unknown changed at a time, the rest at base values:

UNKNOWN (RANGE)	RESULT
Current toward the gap (0.1→0.3 m/s)	held 100% → 76%
Downflow behind the boom (15→50%)	kept 100% → 88%
Bag fabric leak (10→40%)	kept 100% → 88%
Swim speed (5→20 cm/s)	held 100% → 99%
Gap width (500→1,200 m)	226 → 541 kW (base 436)
Water depth (6→12 m)	361 → 484 kW

So the pilot measures current, downflow and bag leak first. Chart in the GitHub README.

HOW WE WOULD TEST AND DEMONSTRATE IT IF SHORTLISTED

From simulation to sea, one measured step at a time

Done

1 · Simulation

Real-data planner, curtain, boom, release and logistics models; online dashboard; stress test; 25 tests.

Months 1–2

2 · ENEC data

Alert history, screen-clearing logs, intake flow, depth survey and species ID: recalibrate the drift, curtain and release models on the real site.

With ENEC

3 · Sea pilot

One curtain section and one uncrewed boom trial in bloom season; FANR approval; drift checked against tracked swarms; survival and effects on other species.

Then

4 · Scale-up

Full gap; live ENEC alerts into the planner; learning model trained on camera and swarm records corrects the forecast.

Pass marks

- Pilot: ≥95% of jellyfish held at the curtain section at the measured current
- Pilot: bag keeps ≥90% at 0.2 m/s; ≥90% still alive 24 h after release
- No harm to non-target species; fewer screen-clearing hours

Open work and AI use

Live dashboard: enechackathon.streamlit.app · code, data, 25 tests: github.com/Shaaha-7/ENEC. AI assistants helped search literature, check claims, draft text and write code; an open text-to-speech model voices the video. The team reviewed every decision and figure.

Sources: Aravind 2024 (UAEU) · SeaLifeBase · J. Exp. Biol. 2018 · FRDC 1999/138 · Bulson 1961, 1968 · UVM Sea Grant · Copernicus SMOC, ERA5 · Gulf News, Nat. Geo. 2013, Vice 2021 · IEEE Spectrum · Jamal et al., PAAET Kuwait · Lo 1991 (bubble barriers) · Hydrotechnik Lübeck. Links, full tables and charts: GitHub README.

Approvals we plan for

FANR for any change at the intake; EAD for marine works and moving live animals; Coast Guard and plant security for boats; an engineering safety case for air near the pumps.

What we need from ENEC

Screen-clearing logs and alert history; intake flow rate; a depth survey of the channel mouth; species ID at Barakah; event frequency.

Beyond the intake

Survey polyps on the breakwater and intake structures, where blooms start. The same kit fits Gulf desalination plants (Taweelah, Fujairah, Oman) and other coastal power stations.